

Flow behavior in c-d nozzles

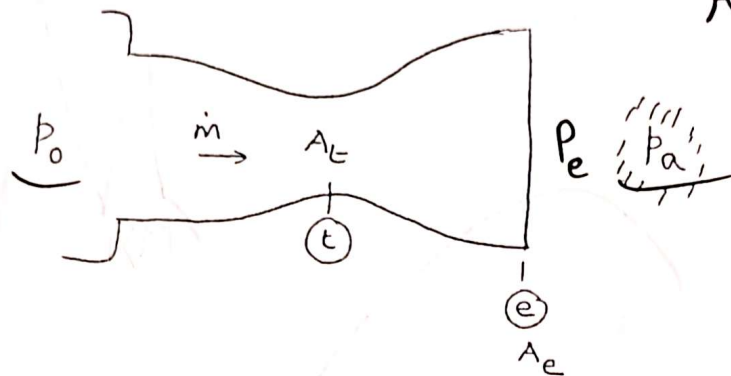
The Mach number at the exit of a c-d nozzle is governed by two factors

① the difference between the pressure in the regions just upstream and downstream of the nozzle; more precisely, the ratio of the pressures

$$\frac{P_e}{P_o}$$

② the nozzle area ratio

$$\frac{A_e}{A^*} \begin{cases} M_e < 1 \\ M_e > 1 \end{cases}$$



In the figure, the c-d nozzle is between regions with pressures P_o and P_a , and has area ratio $\frac{A_e}{A_t}$. Thus, the values of (P_o/P_a) and (A_e/A_t) govern the Mach number at the nozzle exit (M_e).

$$\frac{P_o}{P_a}$$

* If $\left(\frac{P_o}{P_a}\right)$ is "low," then the flow through the nozzle will be entirely subsonic, and both M_t and M_e will be less than 1. The highest Mach number will occur at the throat. Since $M_e < 1$, $P_e = P_a$.

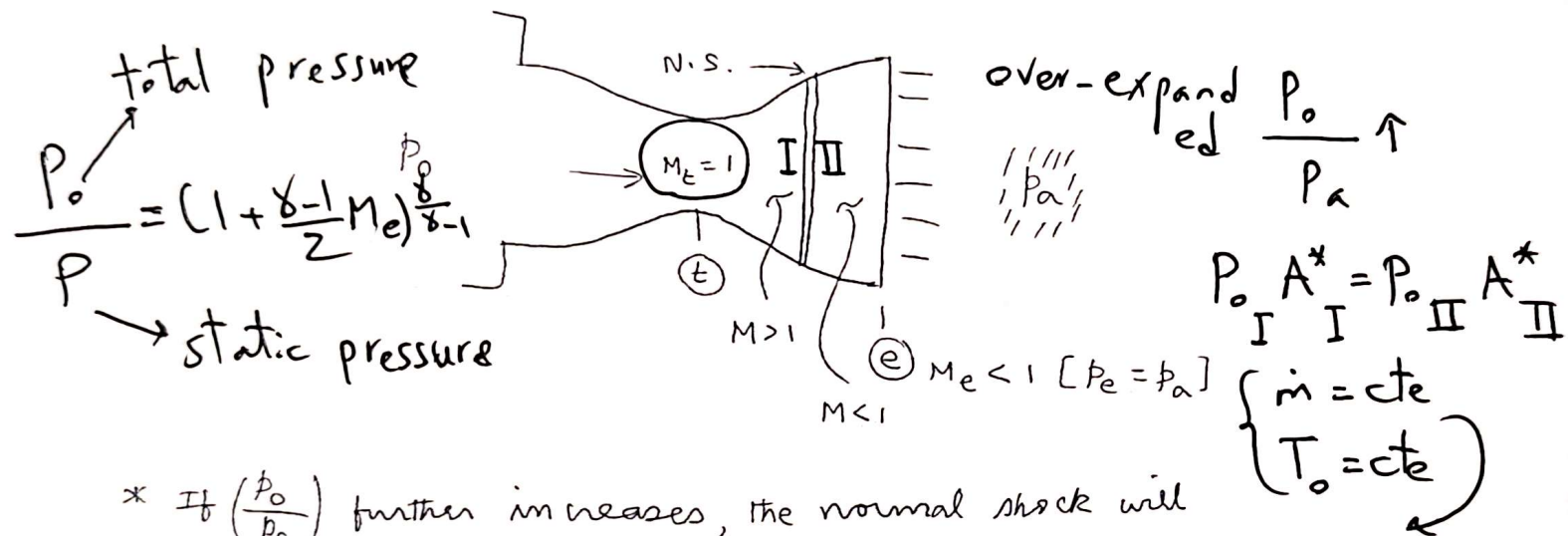
$$\frac{P_o}{P_a} \downarrow$$

$$\begin{matrix} P_o & M_t < 1 & M_e < 1 & P_a \\ & \uparrow & & \\ & \text{highest} & & \end{matrix}$$

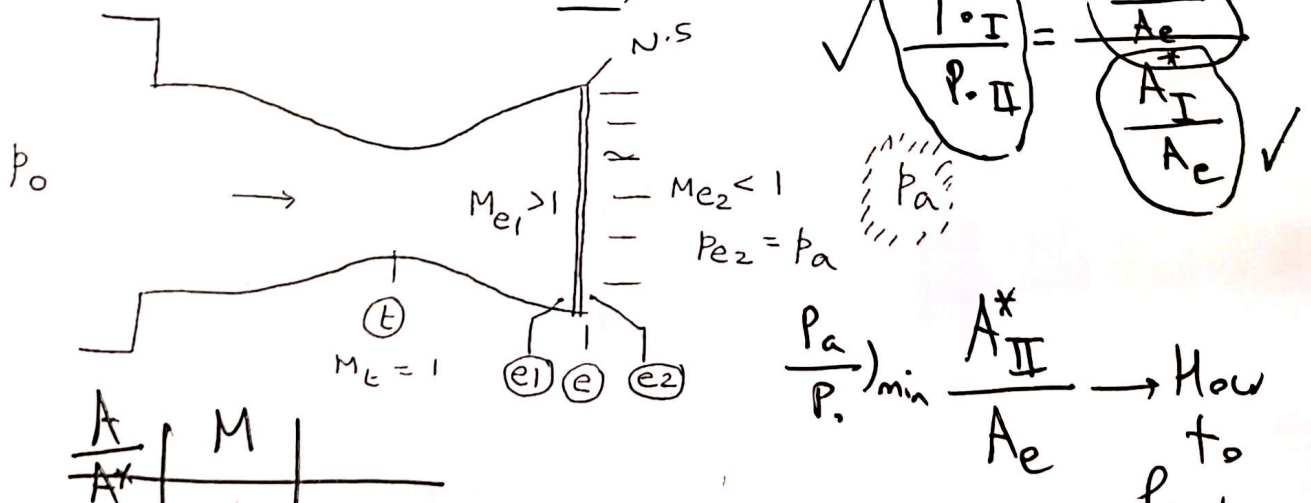


* If $\left(\frac{P_0}{P_a}\right)$ is larger, $M_t = 1$, and the flow will accelerate to supersonic velocities in the diverging portion of the nozzle before encountering a normal shock. The flow then becomes subsonic and $M_e < 1$.

$$\frac{P_a}{P_0} \text{ max}$$



* If $\left(\frac{P_0}{P_a}\right)$ further increases, the normal shock will move downstream and will station itself at the exit plane at a specific $\left(\frac{P_0}{P_a}\right)$ value.



Area-M relation

$\frac{A_e}{A^*} = f(M_e) \begin{cases} M_e > 1 \\ M_e < 1 \end{cases}$

The value of $\left(\frac{p_0}{p_a}\right)$ at this condition is, say, $\left(\frac{p_0}{p_a}\right)_{crit}$

and will be a function of nozzle area ratio $\left(\frac{A_e}{A_t}\right)$.

* If $\left(\frac{p_0}{p_a}\right)$ is increased further, the shock will move

outside the nozzle, and $M_e > 1$. The flow will continuously accelerate from $M < 1$ at the upstream reservoir, reach $M_t = 1$ at the throat, and leave the nozzle at $M_e > 1$.

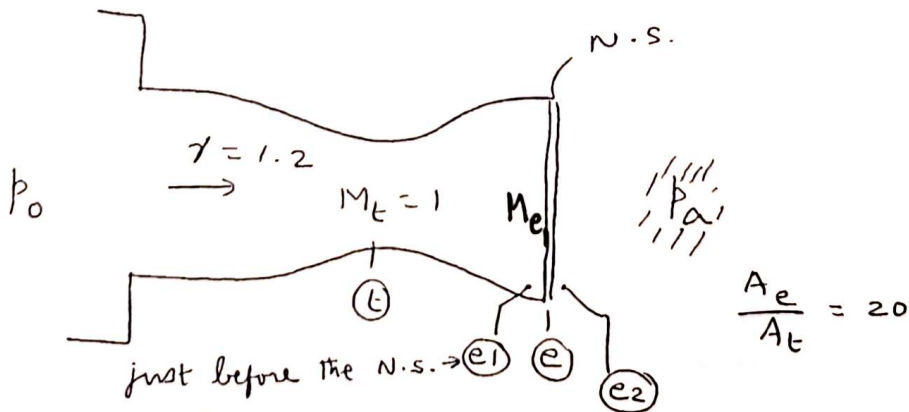
* Thus the minimum $\left(\frac{p_0}{p_a}\right)$ required to have a shock-free flow in the c-d nozzle is $\left(\frac{p_0}{p_a}\right)_{crit}$.

Example

1.2 A c-d nozzle has $\frac{A_e}{A_t} = 20$. Find $\left(\frac{p_0}{p_a}\right)_{\text{crit}}$.

$\frac{A}{A^*}$	M
20	$0 > 1$

Assume that the nozzle is operating with $\left(\frac{p_0}{p_a}\right) = \left(\frac{p_0}{p_a}\right)_{crit}$



$$A_{e1} = A_e \Rightarrow \frac{A_{e1}}{A_t} = 20$$

$$\frac{A_{e1}}{A^*} = \frac{A_{e1}}{A_t} = 20$$

$M_t = 1$

$$\frac{A_{e1}}{A^*} = \frac{1}{M_{e1}} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M_{e1}^2 \right) \right]^{\frac{\gamma+1}{2(\gamma-1)}}$$

Solve for M_e by trial-and-error, using MATLAB or online gas dynamic calculator. Observe $M_{e1} > 1$.

$$M_{e1} = 3.76 \quad (\gamma=1.2) \quad \text{is en, no work} \quad M_{e1} = 3.76$$

$$p_{e1} = \frac{p_{0e1}}{\left(1 + \frac{\gamma-1}{2} M_{e1}^2 \right)^{\frac{\gamma}{\gamma-1}}} = \frac{p_{0e1}}{197.78} = \frac{p_0}{197.78} \quad p_{e1} \quad M_{e2} \quad p_{e2}$$

Use the normal shock relation,

$$\frac{p_{e2}}{p_{e1}} = \frac{2\gamma}{\gamma+1} M_{e1}^2 - \frac{\gamma-1}{\gamma+1}, \quad \text{N.S relation}$$

online normal shock calculator to get

$$\frac{p_{e2}}{p_{e1}} = 15.33 \quad (M_{e1} = 3.76, \gamma = 1.2)$$

$$p_{e2} = 15.33 p_{e1} = 15.33 \cdot \frac{p_0}{197.78} = 0.0775 p_0$$



$\frac{A}{A^*}$	M	$\frac{P}{P_a}$
☐	$M < 1$	☒
☐	$M > 1$	☐

Shocks at the throat

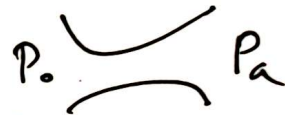
$$M_{e2} < 1 \Rightarrow p_{e2} = p_a$$

$$p_a = 0.0775 p_0 \Rightarrow \frac{p_0}{p_a} = \frac{1}{0.0775} = 12.90$$

This is the $\left(\frac{p_0}{p_a}\right)_{crit}$ value: $\left(\frac{p_0}{p_a}\right)_{crit} = 12.90$

Shock at the end

Observe that this result depends on the nozzle area ratio; 20 in this case. If $\left(\frac{A_e}{A_t}\right)$ had been different, $\left(\frac{p_0}{p_a}\right)_{crit}$ would have been different as well.



Example: An ideal rocket thrust chamber has $p_0 = 1 \text{ MPa}$ and nozzle area ratio of $\overset{Ex}{1.2}$. Will the nozzle be shock-free during operation at 10-km altitude? What are the exit Mach number and the exit pressure? $\gamma = 1.2$.

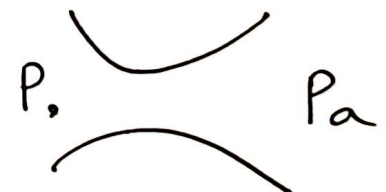
$$\text{at 10-km altitude, } p_a = 26.5 \text{ kPa}$$

$$p_0 = 1 \text{ MPa} = 1000 \text{ kPa}$$

$$\left(\frac{p_0}{p_a}\right)_{actual} = \frac{1000}{26.5} = 37.74$$

The nozzle area ratio = 20, for which $\left(\frac{p_0}{p_a}\right)_{crit} = 12.90$

$$\left(\frac{p_0}{p_a}\right)_{actual} > \left(\frac{p_0}{p_a}\right)_{crit} \Rightarrow \text{shock free nozzle flow.}$$



The exit Mach number is fixed by area ratio (20)

This was already found. $M_e = 3.76$

$$p_e = \frac{p_{0e}}{\left(1 + \frac{\gamma-1}{2} M_e^2\right)^{\frac{\gamma}{\gamma-1}}} = \frac{p_{0e}}{\left(1 + \frac{1.4-1}{2} (3.76)^2\right)^{\frac{1.4}{1.4-1}}} = 5056 \text{ Pa} = 5.056 \text{ kPa}$$

$$p_a = 26.5 \text{ kPa}$$

$p_e < p_a$ (overexpanded jet)

$$M_e \left(\frac{p_0}{p_a} \right)_{act}$$

$$\left(\frac{p_0}{p_a} \right)_{crit}$$

$$M_e = 3.76$$

$$p_e = 5.05$$

$$p_a = 26.5$$

over-expanded

Example: What is the lowest altitude where the thrust chamber of the previous example operate shock free?

$$\left(\frac{p_0}{p_a} \right)_{crit} = 12.90$$

$$p = 1 \text{ MPa}$$

$$p_0 = 1 \text{ MPa} \Rightarrow \left(\frac{p_0}{p_a} \right)_{crit} = \frac{10^6}{12.90} = 77519 \text{ Pa}$$

From atmospheric table (Table 1, p 700) and interpolating,

$$h_{crit} = 2211 \text{ m} \approx 2.2 \text{ km}$$

The lowest altitude for shock-free operation is 2.2 km.

At $h < h_{crit}$, there will be a shock in the diverging portion of the c-d nozzle.

Shock free
2.2 km
shock in nozzle

Comment

Observe that, for nozzle flow with $\left(\frac{p_0}{p_a}\right)_{\text{actual}}$ just greater than

$\left(\frac{p_0}{p_a}\right)_{\text{crit}}$, $p_e < p_a$. To attain the fully expanded

condition ($p_e = p_a$), $\left(\frac{p_0}{p_a}\right)$ has to be further increased.

Thus, if the flow is stated to be fully expanded or

underexpanded ($p_e > p_a$), it follows that $\left(\frac{p_0}{p_a}\right)_{\text{actual}} > \left(\frac{p_0}{p_a}\right)_{\text{crit}}$,

and there can be no shocks in the nozzle.

$$\frac{p_0}{p_{a \text{ act}}} > \frac{p_0}{p_{a \text{ crit}}}$$

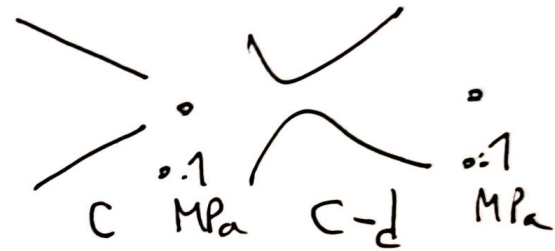
Shock-free

$$\bar{M} = 25 \quad \gamma = 1.2 \quad p_0 = 1 \quad T_0 = 3000 \text{ K} \quad A_t = 0.1 \text{ m}^2 \quad Me$$

Example: Plm 11.2

An ideal rocket thrust chamber has $p_0 = 10 \text{ MPa}$ and $T_0 = 3000 \text{ K}$. $\gamma = 1.2$, $\bar{M} = 25$, $A_t = 0.1 \text{ m}^2$. Calculate Me , u_e , p_e and $\dot{m}$, and c^* and C_f for each of the following cases:

- ① converging nozzle, $p_a = 0$
- ② c-d nozzle ($A_e = 4.06 \text{ m}^2$), $p_a = 0$
- ③ converging nozzle, $p_a = 0.1 \text{ MPa}$
- ④ c-d nozzle ($A_e = 4.06 \text{ m}^2$), $p_a = 0.1 \text{ MPa}$



$$R = \frac{\bar{R}}{\bar{M}} = \frac{8314.3}{25} = 332.6 \frac{\text{J}}{\text{kg} \cdot \text{K}}, \quad \gamma = 1.2$$

Case ①



②
Conv. noz: $\frac{A_e}{A_t} = 1$

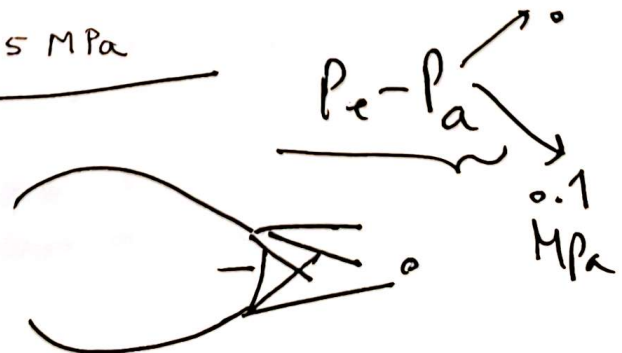
Let $p_e = p_a = 0$
 $p_{0e} = p_0 = 10 \text{ MPa}$
 isen $\Rightarrow \frac{p_{0e}}{p_e} \rightarrow \infty$
 $(1 + \frac{\gamma-1}{2} M_e^2)^{\frac{\gamma}{\gamma-1}}$

$$\left(1 + \frac{\gamma-1}{2} M_e^2\right)^{\frac{\gamma}{\gamma-1}} \rightarrow \infty \Rightarrow M_e \rightarrow \infty$$

$M_e \nless 1 \Rightarrow M_e = 1$
 conv. noz.

$$p_e = \frac{p_{0e}}{\left(1 + \frac{\gamma-1}{2} M_e^2\right)^{\frac{\gamma}{\gamma-1}}} = 5.645 \text{ MPa}$$

$p_a = 0.1 \text{ MPa}$



$$T_e = \frac{T_{0e}}{1 + \frac{\gamma-1}{2} M_{e\leftarrow}^2} = \underline{2727 \text{ K}} \left\{ T_{0e} = T_0 = 3000 \text{ K} \right\}_{\text{isen}}$$

$$u_e = M_e \sqrt{\gamma R T_e} = \underline{1043 \text{ m/s}}$$

$$\rho_e = \frac{p_e}{R T_e} = \frac{5.645 (10^6)}{(332.6)(2727)} = \underline{6.224 \frac{\text{kg}}{\text{m}^3}}$$

$$A_e = A_t = 0.1 \text{ m}^2$$

$$\dot{m} = \rho_e u_e A_e = 649.2 \text{ kg/s} \checkmark$$

$$\tau_{\text{conv}} = \dot{m} u_e + (p_e - p_a) A_e$$

$\begin{matrix} \nwarrow 649.2 & \nwarrow 1043 \text{ m/s} & \nwarrow 0.1 \\ \nearrow 5.645 & \nearrow p_a & \nearrow 0 \end{matrix}$

$$= (0.6771 + 0.5645) = \underline{1.242 \text{ MN}}$$

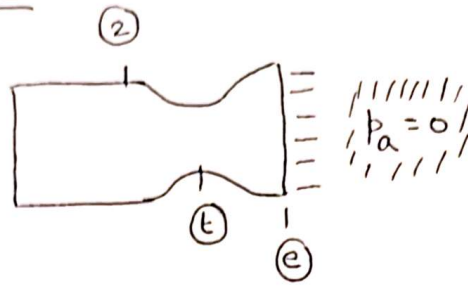
$$C^* = \frac{p_0 A_t}{\dot{m}} = \frac{(10)(10^6)(0.1)}{649.2} = \underline{1540 \text{ m/s}}$$

$$\left\{ \text{From Egn (11.9), } C^* = \underline{1540 \text{ m/s}} \right\}$$

$$C_{\tau_{\text{conv}}} = \frac{\tau_{\text{conv}}}{p_0 A_t} = \frac{1.242}{(10)(0.1)} = \underline{1.242}$$

$$\left\{ \text{From Egn (11.12), } C_{\tau_{\text{conv}}} = \underline{1.24, p_a = 0} \right\}$$

Case (2)



$$A_e = 4.06 \text{ m}^2$$

$$\frac{A_e}{A_t} = \frac{4.06}{0.1} = 40.6$$

$$\frac{A_e}{A^*} = \frac{A_e}{A_t} = 40.6$$

$M_e = 1$

$$\frac{A_e}{A^*} = \frac{1}{M_e} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M_e^2 \right) \right]^{\frac{\gamma+1}{2(\gamma-1)}}$$

Trial - and - error, or online calculator $\Rightarrow M_e = 4.25$

$$p_e = \frac{p_{0e} \xleftarrow{\text{isen}} p_0 = 10 \text{ MPa}}{\left(1 + \frac{\gamma-1}{2} M_e^2 \right)^{\frac{\gamma}{\gamma-1}}} = 0.020476 \text{ MPa}$$

$$T_e = \frac{T_{0e} \xleftarrow{\text{isen}} T_0 = 3000 \text{ K}}{1 + \frac{\gamma-1}{2} M_e^2} = 1069 \text{ K}$$

$$u_e = M_e \sqrt{\gamma R T_e} = 2776 \text{ m/s}$$

$$\rho_e = \frac{p_e}{R T_e} = \frac{0.020476 (10^6)}{(332.6)(1069)} = 0.05759 \frac{\text{kg}}{\text{m}^3}$$

$$\dot{m} = \rho_e u_e A_e = 649.1 \text{ kg/s}$$

$$c^* = \frac{p_0 A_t}{\dot{m}} = \frac{10 (10^6) (0.1)}{649.1} = 1540 \text{ m/s}$$

$$(11.9) \Rightarrow c^* = 1540 \text{ m/s}$$

$$\tau = \dot{m} u_e + (p_e - p_a) A_e$$

$$= 1.802 + 0.083 = \underline{1.885 \text{ MN}}$$

$$C_T = \frac{\tau}{p_0 A_t} = \frac{1.885}{(10)(0.1)} = \underline{1.885}$$

$$\frac{C_T}{C_{T_{\text{conv}}}} > 1$$

$$(11.11) \Rightarrow C_T = 1.885$$

$$\frac{C_T}{C_{T_{\text{conv}}}} = \frac{1.885}{1.242} = \underline{1.52} \left\{ \begin{array}{l} \text{increase in thrust} \\ \text{due to divergence} \\ (p_a/p_0) = (0/10) = \underline{\underline{0}} \end{array} \right.$$

Case ③

$$\text{conv. noz.} \therefore p_a = \underline{0.1 \text{ MPa}}, p_0 = 10 \text{ MPa}$$

$$\frac{0.1}{10} = 0.01$$

$$p_{0e} = p_0 = 10 \text{ MPa}.$$

isem

$$\text{Let } p_e = p_a = 0.1 \text{ MPa}$$

$$\left. \begin{array}{l} p_{0e} = p_0 = 10 \text{ MPa} \\ \text{isem} \\ \text{Let } p_e = p_a = 0.1 \text{ MPa} \end{array} \right\} \Rightarrow \frac{p_{0e}}{p_e} = \frac{10}{0.1} = 100$$

$$\left(1 + \frac{\gamma-1}{2} M_e^2\right)^{\frac{\gamma}{\gamma-1}} = 100$$

$$\Rightarrow M_e = \underline{\underline{3.40}}$$

$$M_e \nless 1 \Rightarrow \boxed{M_e = 1}$$

All the exit conditions are the same as in case ①.

$$\dot{m} = 649.2 \text{ kg/s}, p_e = 5.645 \text{ MPa}, u_e = \underline{1043 \text{ m/s}}$$

$$\tau = \dot{m} u_e + (p_e - p_a) A_e = 0.6771 + 0.5545 = \underline{1.23 \text{ MN}}$$

$$c^* = 1540 \text{ m/s}, \quad c_{T_{\text{conv}}} = \frac{\tau}{p_0 A_t} = \frac{1.23}{(10)(0.1)} = 1.23$$

$$(11.9) \Rightarrow c^* = 1540 \text{ m/s}, \quad (11.12) \Rightarrow c_{T_{\text{conv}}} = 1.23$$

Case (4)

$$A_e = 4.06 \text{ m}^2, \quad \frac{A_e}{A_t} = 40.6 \Rightarrow M_e = 4.25$$

All exit conditions (M_e , u_e , p_e and $\dot{m}$) are the same as in case (2)

$$p_e = 0.020476 \text{ MPa}$$

$$c^* = 1540 \text{ m/s}$$

$$(11.9) \Rightarrow c^* = 1540 \text{ m/s}$$

$$A_e = A_t \rightarrow 1.83$$

$$\tau = \dot{m} u_e + (p_e - p_a) A_e$$

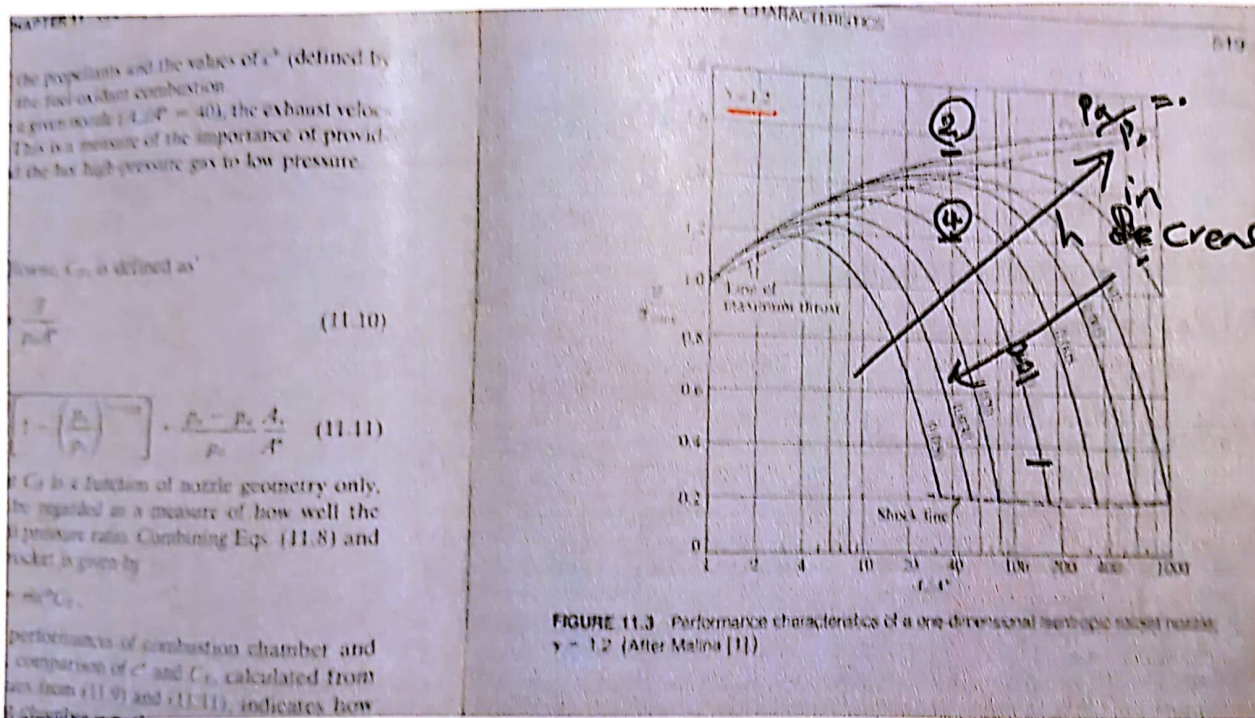
$$= 1.802 - 0.323 = 1.479 \text{ MN}$$

$$A_e = 40 A_t \quad (2), (4)$$

$$c_{\tau} = \frac{\tau}{p_0 A_t} = \frac{1.479}{(10)(0.1)} = 1.479$$

$$(11.11) \Rightarrow c_{\tau} = 1.479$$

$$\frac{C_T}{C_{T_{CONV}}} = \frac{1.479}{1.23} = 1.20 \quad \left\{ \begin{array}{l} \text{increase in thrust} \\ \text{due to divergence} \\ (p_a/p_0) = (0.1/10) = 0.01 \end{array} \right.$$



Compare the results with those depicted in Figure 11.3, text.